Solvency Field Theory V13.1-V3

From Locally Real Writes to the Selected Einstein-Form Gravity Branch

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CURRENT STATUS

P1, P2, and P3 remain discharged for the selected regular local gravity branch. V13.1-V3 derives the structural coupling relation G = a_phi c^3 / hbar and the selected Einstein-form response. The later electron closure/operator program supplies a comparison-separated, zero-continuous-parameter conditional numerical prediction under the explicit minimal optical-normalization postulate c = 1. The full first-principles numerical derivation remains open by that one universal dimensionless representation normalization.

POST-CANON NUMERICAL STATUS
Conditional identity prediction (c = 1)
G0 = 6.674306513706680 x 10^-11 m^3 kg^-1 s^-2
0.976 ppm above the 2022 CODATA central value; present relative standard uncertainty is 22.5 ppm.
Claim boundary: conditional prediction, not a completed first-principles derivation.

1. What this paper claims

Starting from the frozen locally real particle and irreversible-write ontology, V13.1-V3 constructs a minimal symmetric settlement carrier, a boundary-complete conserved write-momentum current, and a natural ordered update law. It derives a smooth capacity-normalized continuum representation, selects the regular local metric state, and closes the selected field equation

$$G_{\mu\nu}[g(\Sigma)] = \frac{8\pi G_{\mathrm{eff}}}{c^4}\, \mathcal{W}_{\mu\nu}$$

The original equation was frozen before the hostile gravity program and was not retuned. The later microscopic audit discharged the three premises originally attached to that freeze for the selected branch. The later numerical-G companion does not alter this field equation or its structural claim boundary.

2. Native chain

Real support spends one invariant causal budget c.

Massive particles are persistent closed support with definite local orientation and cadence.

Completed writes are irreversible and permanently ordered.

The frozen scalar, mixed, trace, and STF gravity sectors assemble into a symmetric rank-two carrier under local frame changes.

Four-momentum transport through oriented support gives a rank-two source current.

Completed vertex momentum balance cancels internal endpoint defects in the worldline/edge-supported current.

Exact updates are natural on physical relabeling-equivalence classes.

3. P1 - composition and continuum

Exact histories are finite ordered products; they are not assumed to be exponentials. In a macroscopic capacity-normalized sector, bounded convergent event ensembles admit the ordered continuum representation U = P exp(integral X ds), without subdividing one physical event.

4. P2 - address naturality

A physical write is relational. Coordinates, event names, raw phase zeros, and port labels are not physical inputs. Exact admissibility and update commute with relabeling. Tensor-valued distributions provide the coordinate-free continuum bridge, and the selected local kernel inherits covariance by descent.

5. P3 - sufficient local state

The selected regular local gravity state is constrained/gauge-reduced (h_ij, K_ij), current source projections, complete current matter/carrier state, and physical topology/boundary metadata. Independent torsion, higher-rate, memory, and BCH registers are not present in this branch. The response is a same-write settlement update followed by the regular metric congruence update.

6. Source-current repair

An isolated point tensor is not generally divergence-free. The conserved source is the complete oriented worldline/edge current. Completed vertices cancel internal endpoint defects; boundary terms remain until the full physical current is included. This repair supersedes the earlier point-atom conservation sentence.

7. What survived the hostile stack

Without changing the field equation, the original V3 program obtained or conditionally recovered:

- Newtonian exterior and interior limits; full weak light bending and Shapiro delay;
- Schwarzschild and local Birkhoff behavior; one-metric lensing and dynamics;
- frame dragging and exact Kerr existence; two TT wave modes at c;
- leading quadrupole radiation and pulsar regressions;
- TOV stars, binding, stability, and tidal response for external EOS models;
- homogeneous collapse, trapped surfaces, linear Schwarzschild ringdown, FLRW compatibility, and the trace-capacity bridge.

The full post-Newtonian hierarchy, strong-field uniqueness/stability, matter microphysics, horizon mechanics, and cosmological write dynamics remain open.

8. Structural root relation

$$G_{\mathrm{eff}} = \frac{a_\phi c^3}{\hbar}$$

The V13.1-V3 equations are invariant under a common rescaling of the root length. Compactness, mass ratios, closure laws, capacity ratios, and horizon/Compton comparisons do not by themselves fix that absolute scale. The original canon therefore correctly left numerical G open within its own scale-free stack.

9. Later electron-to-root bridge

The later locally real electron program adds a dimensionless bridge that was absent from V13.1-V3. It derives the six-sector register baseline, the limiting electron recurrence, and the occupied-state availability quotient. Under the explicit minimal optical identity representation c = 1, the conditional formula is

$$\Gamma_e^{(0)} = \left(1 - \frac{q_*}{3}\right)\left(\frac{e^{3L_{\mathrm{eff}}}}{2\pi}\right)^2, \qquad G_0 = \left(\varepsilon_e^{(0)}\right)^2 \frac{\hbar c}{m_e^2}$$

$$G_0 = 6.674306513706680 \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$$

This is a zero-continuous-parameter conditional prediction. It is not promoted to a complete first-principles derivation because the current operator stack still permits the universal family Gamma = c chi_e B6 for any positive representation normalization c. Identity c = 1 is the minimal postulate; it is not yet an independent theorem.

10. Threshold-recurrence provenance

The key phase decimal is not an independent input. At the exact matter/light boundary, P_coh(q, pi) u = 1/3 eliminates u from the recurrence and gives

$$q = \exp\left[-\frac{3\pi}{4} P_{\mathrm{coh}}(q, \pi) - \frac{L_{\mathrm{eff}}}{12}\right], \quad q_* = 0.091710909655065\ldots$$

The unique stable root is solved first. Only afterward are u_* and the charge-conjugate weak-lane edges reconstructed, giving |beta - pi/3| = 0.021432753081800... . The earlier ...1799 decimal is corrected at the fifteenth rounded decimal place. The correction changes numerical G only at approximately 10^-13 relative because the production calculation uses q_* directly.

Receipt / date	Frozen result	Chronology role
MWO-G11B / 2026-07-22	Recurrence q = exp[-K P_coh]	Predates final residual formula
MWO-G11D / 2026-07-22	K = 3pi/4 + (L_eff/4)u	Predates threshold edge
UNI-G13L10E / 2026-07-23	Matter/light threshold P_coh u = 1/3	Eliminates u without G
UNI-G13L15E / 2026-07-24	All-twelve beta edge	Predates occupied-state quotient
UNI-G13L25E / 2026-07-24	Exact edge; 0.999-edge coincidence rejected	Correction predates G13L39E
UNI-G13L39E / 2026-07-26	chi_e = 1 - q_*/3; final comparison	First successful quotient

The project was not historically ignorant of measured G. The defensible statement is comparison separation: the final candidate formula was generated from frozen internal equations before numerical comparison. This replaces unqualified “blind derivation” language.

11. CODATA central-value drift

The sub-ppm central-value offset must be read against both historical drift and present uncertainty. Official NIST/CODATA central values moved by 68.93 ppm from 2010 to 2018; 2018 and 2022 retained the same central value.

CODATA	Central G (SI)	Std. uncertainty	Relative u (ppm)	G0 - central (ppm)
2010	6.67384e-11	0.00080e-11	119.87	+69.90
2014	6.67408e-11	0.00031e-11	46.45	+33.94
2018	6.67430e-11	0.00015e-11	22.47	+0.976
2022	6.67430e-11	0.00015e-11	22.47	+0.976

Accordingly, SFT claims consistency with the present CODATA interval and a 0.976-ppm central-value offset. It does not claim experimental confirmation at sub-ppm precision.

12. Branch and calibration firewall

- Native structural branch: a_phi and G remain symbolic; no measured G enters the field-equation derivation.
- Identity prediction branch: set the explicit minimal representation c = 1, derive the dimensionless electron share, then use the independently measured inertial electron mass for the SI handoff.
- One-anchor control: use measured G once to determine c; this is calibration and may never be called a prediction of G.
- The identity prediction and one-anchor control remain separate permanently; no sector-specific retuning is allowed.

13. Updated claim boundary

SFT may claim that the selected structural gravity branch and P1-P3 discharge are receipt-backed. It may claim the root relation G = a_phi c^3 / hbar. It may additionally claim the comparison-separated, zero-continuous-parameter conditional numerical prediction under the explicit minimal optical-normalization postulate c = 1.

SFT may not claim that c = 1 has been independently derived, that current experiments confirm the result at one-part-per-million precision, that absolute SI units arise from pure dimensionless axioms, or that the full strong-field, matter, and cosmological programs are complete.

Matter models may fail. Gravity does not move to save them.

14. Program handoff

Further homogeneous stability, profile, formation, and contact searches for the normalization are stopped. The numerical-G derivation reopens only if an independent representation theorem fixes c. The active locally real particle program now moves to the geometry-to-particle atlas, beginning with the anchor-free charged-lepton ratio test:

same charged-lepton geometry + different generation lane -> m_e : m_mu : m_tau

15. External comparison references

Official NIST/CODATA complete and archived constants listings for 2010, 2014, 2018, and 2022; the 2022 electron-mass entry; and SFT V13.1-V3 Numerical-G Companion v1.1 with AUD-G13L42G. Full URLs and receipt hashes are preserved in the companion and selected-receipts pack.